Experiment 11: Map Coloring using Constraint Satisfaction Problem (CSP)

Aim

To write a Python program to solve the Map Coloring problem using the Constraint Satisfaction Problem (CSP) approach.

Procedure

1. Define the map and neighboring regions.
2. Assign available colors.
3. Check whether neighboring regions have different colors.
4. Assign a valid color to each region.
5. Display the final color assignment.

Source Code

main.py

Visualize

Run ▶

```
graph = {
    'A': ['B', 'C'],
    'B': ['A', 'C', 'D'],
    'C': ['A', 'B', 'D'],
    'D': ['B', 'C']
}

colors = ['Red', 'Green', 'Blue']
result = {}

def is_safe(node, color):
    for neighbor in graph[node]:
        if neighbor in result and result[neighbor] == color:
            return False
    return True

def solve(node_list):
    if not node_list:
        return True

    node = node_list[0]

    for color in colors:
        if is_safe(node, color):
            result[node] = color
            if solve(node_list[1:]):
                return True
            result.pop(node)

    return False

solve(list(graph.keys()))

print("Color Assignment:")
for node in result:
    print(node, "->", result[node])
```

Output

```
Color Assignment:
A -> Red
B -> Green
C -> Blue
D -> Red
```

Result

Thus, the Map Coloring problem was solved successfully using CSP.